GGIB Part 5 - Contamination Resistance and Gaming Defenses

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The Threat Model

Three categories of gaming threat, each requiring distinct defenses:

1. **Contamination** - Task instances leak into training data. The model memorizes correct responses rather than demonstrating evidence-sensitive behavior.
2. **Strategy exploitation** - The model learns a surface-level behavioral strategy (always hedge, always revise, match output format) that scores well without processing evidence.
3. **Scoring exploitation** - The model learns the scoring system's structure and optimizes for the score function directly rather than for the underlying epistemic behavior.

Strategy exploitation is the most dangerous for this benchmark specifically. On a performance benchmark, a model cannot fake the right answer. On a metacognition benchmark, a model can fake the right *behavior* because the

behaviors being measured (hedging, refusing, revising) are conversational acts the model has already been trained to produce via RLHF. The defenses must be designed primarily against strategy exploitation, with contamination and scoring exploitation as secondary concerns.

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1. Making Task Instances Novel

1.1 Procedural Generation With Parameterized Templates

Each task family uses a generation schema (defined in GGIB Part 2 - Metacognition Task Family Design) that separates structure from content. The structure defines the evidence condition and expected behavior. The content fills in domain-specific details that can be varied without changing the underlying epistemic task.

Generation parameters that vary independently:

Parameter	Variation Range	Purpose
Domain	Chemistry, physics, economics, geography, logic, scheduling, engineering, medical, cybersecurity, environmental	Prevents domain-specific memorization
Numeric values	Procedurally generated within realistic ranges	Prevents specific-value memorization
Entity names	Randomized (substances, companies, locations, people)	Prevents entity-based pattern matching
Evidence body length	Short (3-4 facts) to long (8-12 facts)	Prevents length-based heuristics
Distractor count	0-4 irrelevant facts included	Prevents distractor-count heuristics

Parameter	Variation Range	Purpose
Question phrasing	3-5 phrasings per question type	Prevents prompt-template matching
Evidence order	Randomized within each instance	Prevents position-based heuristics

1.2 Private Seeds, Public Schemas

The generation schemas are disclosed in the submission documentation (transparency for reviewers). The random seeds used to generate the actual evaluation instances are held private. This means:

- Anyone can inspect the generation logic and verify it produces valid tasks
- No one can reconstruct the specific instances used for scoring
- New evaluation sets can be generated at any time by drawing new seeds

1.3 Benchmark Refresh Cycles

The evaluation set should be regenerated on a regular cycle (recommended: every 3-6 months after initial deployment). Each refresh uses the same generation schemas with new seeds, producing structurally equivalent but content-distinct instances. This prevents long-term contamination even if specific instances leak.

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2. Reducing Contamination Through Procedural Generation

2.1 Why Static Benchmarks Fail

Static benchmarks (fixed question sets published once) are vulnerable to two contamination vectors:

- **Direct leakage**: The questions appear in training data, either through web scraping of benchmark documentation or through user interactions that

include benchmark questions.

- **Indirect leakage**: Models are fine-tuned on benchmark-adjacent data that teaches the specific patterns the benchmark tests.

Procedural generation addresses direct leakage: if each evaluation instance is generated fresh from a seed, no specific instance exists in any training corpus. It partially addresses indirect leakage: the generation schemas are public, but the specific parameter combinations are not.

2.2 Canary Tasks for Leak Detection

Include 10-20 "canary" instances in each evaluation set. These are instances with deliberately unusual parameter combinations that would be extremely unlikely to appear in any training corpus (e.g., a chemistry problem involving a fictitious element with a made-up name, or an economics scenario set in a nonexistent country).

If a model performs anomalously well on canary tasks relative to standard tasks, it suggests the model has been exposed to the evaluation set. This is a post-hoc detection method, not a prevention method, but it provides an audit trail.

2.3 Version-Controlled Instance Families

Each task instance belongs to a "family" of structurally equivalent instances. Family membership is defined by the evidence condition, the expected behavior, and the key evidence elements, not by the specific content. Within a family, instances share the same scoring logic but differ in surface content.

This means a model that memorizes one instance cannot transfer that memorization to a different instance in the same family, because the content differs. But the benchmark can verify that performance is consistent across instances within a family, which is a contamination signal: if a model performs dramatically better on one instance than on structurally equivalent instances, it may have memorized that specific instance.

3. Resisting Models Trained to Output Hedge Language

This is the primary gaming threat. RLHF-trained models already hedge, refuse, and express uncertainty as default behavior. A benchmark that rewards hedging is trivially gamed by any safety-trained model.

3.1 Sufficient-Evidence Control Tasks (Built Into Design)

From GGIB Part 2 - Metacognition Task Family Design, approximately 40% of task instances have "confident commitment" as the correct epistemic response. A model that hedges on everything accumulates penalties on these control tasks:

- EJA = 0 (wrong epistemic judgment: said "unanswerable" when evidence was sufficient)
- CA = 0 (confidence too low for the appropriate range)
- MEI = 0 (listed missing evidence when nothing was actually missing)

The aggregate ECS score for a blanket hedger is structurally capped at approximately 0.35-0.45 across the full benchmark. This is by design.

3.2 Specificity Requirements (RS and MEI Dimensions)

From GGIB Part 3 - Formal Scoring Framework v0.2, Dimension 2 (Reasoning Specificity) now uses a dual-path architecture that requires the model to both reference AND functionally utilize specific evidence elements. Reference alone (mentioning an element without using it in the reasoning chain) scores at most RS = 0.30. Dimension 5 (Missing Element Identification) continues to use reference-presence matching for set comparison.

A model that outputs ANSWERABLE: no with "There may not be enough information to determine the answer" scores RS = 0 and MEI = 0 on that instance. A model that lists evidence elements without using them to drive its epistemic judgment scores at most RS = 0.30. Only models that functionally utilize evidence elements (transforming, comparing, or linking them to the answerability constraint) can score above RS = 0.30. The ECS score for format-correct theater is capped at approximately 0.55 or lower.

3.3 Label-Swapped Controls

For each task family, include instances where the surface features suggest one evidence condition but the actual condition is different:

- **Looks insufficient, is sufficient:** Short evidence body, informal phrasing, but all required facts are present. A hedge-default model will incorrectly refuse.
- **Looks sufficient, is insufficient:** Long evidence body, authoritative sources, formal phrasing, but one critical fact is missing. A commit-default model will incorrectly answer.
- **Looks contradictory, is consistent:** Multiple sources with different wording but no actual factual conflict. A contradiction-flagging model will incorrectly flag.
- **Looks consistent, is contradictory:** Sources that agree on surface but contain a subtle numerical or logical inconsistency.

If a model's EJA accuracy correlates with surface features (length, formality, source type) rather than actual evidence conditions, the label-swapped controls expose this.

Detection method: Compute EJA separately for label-swapped instances and standard instances. If EJA drops significantly on label-swapped instances, the model is using surface heuristics rather than evidence processing.

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4. Detecting Indiscriminate Refusal

4.1 The Indiscriminate Hedging Penalty (IHP)

Compute the model's refusal/hedging rate on sufficient-evidence control tasks specifically:

```
HedgeRate_sufficient = (number of sufficient-evidence tasks
where model hedged or refused)
```

```
                        / (total number of sufficient-evidence
tasks)

IHP = max(0, HedgeRate_sufficient - tau)
```

Where tau is calibrated from the human baseline: set tau at the 75th percentile of human refusal rate on sufficient-evidence tasks. This means the model is allowed to refuse at a rate up to the upper quartile of human refusal behavior before penalties apply.

IHP is a model-level penalty that is reported alongside the per-task ECS scores. It does not modify individual task scores but provides a diagnostic signal that the model's hedging behavior is evidence-insensitive.

4.2 Cross-Condition Consistency Analysis

For each model, compute the distribution of epistemic judgments (answerable/unanswerable, contradiction/no contradiction, etc.) separately for each evidence condition:

- Sufficient evidence tasks
- Insufficient evidence tasks
- Contradictory evidence tasks
- Misleading abundance tasks
- Non-diagnostic new evidence tasks

An evidence-sensitive model should show different judgment distributions across conditions. An indiscriminate model shows the same distribution regardless of condition.

Detection metric: Chi-squared test for independence between evidence condition and epistemic judgment distribution. A non-significant chi-squared indicates the model's judgments are not evidence-sensitive.

5. Detecting Format Compliance Without Evidence Sensitivity

5.1 The Core Problem

A model can learn the benchmark's output format and produce structurally correct responses without processing the evidence. This is the deepest form of epistemic theater: the model has learned that "when the format asks for ANSWERABLE, CONFIDENCE, REASONING, MISSING, the correct pattern is..."

5.2 The Theater Detection Score (TDS)

From GGIB Part 3 - Formal Scoring Framework, the TDS = EJA - RS metric directly targets this. A model with high EJA (correct categorical judgments) but low RS (non-specific reasoning) is producing format-correct theater. TDS > 0.2 flags this pattern.

5.3 Evidence-Element Perturbation Tests

For a subset of task instances, create perturbation variants where:

- One key evidence element is changed (different number, different substance, different location) but the evidence condition remains the same
- The expected behavior is identical, but the specific evidence elements that should appear in the reasoning are different

If the model's REASONING field is identical or near-identical across perturbation variants (because it is producing template reasoning rather than evidence-specific reasoning), the perturbation test detects this.

Detection method: Compute text similarity (e.g., ROUGE-L or cosine similarity of sentence embeddings) between the model's REASONING responses on perturbation pairs. High similarity (> 0.85) on pairs with different evidence content suggests template reasoning.

5.4 Adversarial Baseline Battery

Before deployment, run the benchmark against seven adversarial baselines. Each baseline implements a single strategy and should score poorly in distinct, predictable ways:

Baseline	Strategy	Expected Failure Pattern
Always-Answer	Answers every task with high confidence	Fails on insufficient/contradictory tasks (EJA = 0, CA = 0, high CD)
Always-Refuse	Refuses every task	Fails on sufficient-evidence controls (EJA = 0, high IHP)
Always-Conditional	Gives conditional answers on everything	Fails on sufficient-evidence tasks and clear insufficient tasks
Always-Revise	Revises whenever new evidence appears	Fails on confirming and non-diagnostic evidence in SEI (RA low)
Never-Revise	Never revises regardless of new evidence	Fails on disconfirming evidence in SEI (RA low)
Keyword-Hedge	Outputs hedge language on every task	Fails on RS (no evidence utilization in reasoning), fails on sufficient controls
Random-Support	Cites random evidence IDs in support fields	Fails on RS (wrong elements referenced, no functional utilization) and MEI (wrong missing elements)

Validation criterion: Each adversarial baseline must score below 0.40 ECS overall. If any baseline scores above 0.40, the benchmark has a gaming vulnerability that must be addressed before deployment.

5.5 Surface Feature Predictability Audit

Before deployment, run a logistic regression predicting the correct epistemic response from surface features only:

- Task text length
- Number of evidence elements
- Domain category
- Presence of authority cues (journal names, titles)

- Presence of numeric values
- Number of sources cited

If the regression achieves accuracy above 60% (well above the ~40%/35%/25% base rate distribution), surface features are leaking the correct response and the generation schema needs revision.

5.6 CCED Surface-Matched Adversarial Pairs (added per C. Brooks full review)

CCED (Confidence Calibration Under Evidence Degradation) is the most exposed family to mimicry because confidence can be learned as a surface-feature mapping without evidence processing. To harden CCED specifically:

At least 20% of CCED instances are **surface-matched adversarial pairs** (see GGIB Part 2 - Metacognition Task Family Design v0.3, CCED Section I for full specification). Three pair types:

- **Pair A:** Identical surface features, different evidence quality. Tests whether confidence tracks evidence or surface cues.
- **Pair B:** Opposite surface features, identical evidence quality. Tests whether confidence tracks formality or evidence.
- **Pair C:** Identical evidence quality, different claim truth. Tests whether confidence tracks claim truth or evidence.

Validation criterion: Compute within-pair confidence deltas. If deltas are large on Pair B or C (where evidence quality is constant), the model is using the wrong signal.

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6. Pre-Deployment Adversarial Validation Checklist

Before the benchmark is used for any evaluation:

- [ ] All seven adversarial baselines score below 0.40 ECS
- [ ] Surface feature regression accuracy is below 60%
- [ ] Label-swapped controls show no significant EJA difference from standard instances for the ideal epistemic agent
- [ ] Canary tasks are included and their performance is logged for post-hoc leak detection
- [ ] Sufficient-evidence controls comprise at least 35% of all task instances
- [ ] The IHP threshold (tau) is calibrated from human baseline data
- [ ] Perturbation variants exist for at least 20% of task instances
- [ ] The TDS metric has been validated on worked examples (see GGIB Part 3 - Formal Scoring Framework)
- [ ] CCED surface-matched adversarial pairs comprise at least 20% of CCED instances
- [ ] Generation seeds for the evaluation set are stored securely and not included in any public documentation

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Summary: Defense Layering

Threat	Primary Defense	Secondary Defense	Detection Method
Contamination (memorization)	Procedural generation with private seeds	Canary tasks, benchmark refresh cycles	Cross-instance consistency within families
Blanket hedging	Sufficient-evidence controls (~40% of tasks)	IHP penalty	HedgeRate on control tasks
Epistemic theater (format	RS and MEI specificity	TDS diagnostic	TDS > 0.2 flags theater

Threat	Primary Defense	Secondary Defense	Detection Method
compliance)	requirements		
Surface heuristics	Label-swapped controls	Surface feature predictability audit	EJA drop on label-swapped tasks
Template reasoning	Evidence-element perturbation tests	Cross-perturbation REASONING similarity	ROUGE-L > 0.85 on perturbation pairs
Indiscriminate refusal	Sufficient-evidence controls	Cross-condition consistency analysis	Chi-squared test for evidence sensitivity
Performative revision	Non-diagnostic evidence conditions in SEI	Confirming evidence conditions	RA breakdown by evidence type
Confidence mimicry (CCED)	Surface-matched adversarial pairs	Prior knowledge contamination checks	Within-pair confidence delta analysis

No single defense is sufficient. The defenses are layered so that defeating one does not defeat the others. A model that passes all eight defense layers is demonstrating behavior that is, at minimum, very difficult to produce without evidence-sensitive processing.

Related Notes

- AGI-Benchmark-Blueprint
- GGIB-Part1-Critical-Analysis

- GGIB Part 2 - Metacognition Task Family Design
- GGIB Part 3 - Formal Scoring Framework
- GGIB Part 4 - Human Baseline Design
- Orchestrator_Epistemic_Capture_v2
- Metacognition-Benchmark-Critique
- Metacognition-Benchmark-Synthesis-Report

Tags

#AGI #benchmark #metacognition #contamination-resistance #gaming-defense #adversarial-validation #GGIB #Kaggle #competition #epistemic-theater